

ATLAS

Predictive Traffic Command for Bengaluru

A detailed submission dossier explaining the prototype, data flow, algorithms, impact, scalability, limitations, and demo narrative.

Theme	Event-Driven Congestion (Planned & Unplanned)
Prototype	Local full-stack web app + ML/analytics backend
Data	Astram anonymized Bengaluru incident history
Core claim	Turns raw incident logs into prediction, replay, and deployment decisions

ATLAS - Predictive Traffic Command for Bengaluru

A submission dossier for reviewers, judges, and teammates

Theme: Event-Driven Congestion, Planned & Unplanned **Prototype:** Local full-stack web app + ML/analytics backend
Data source: Astram anonymized event dataset, Bengaluru traffic incidents **Core idea:** Convert a messy city incident log into a live decision-support system for congestion prediction, cascade detection, police deployment, barricading, diversions, and post-event learning.

1. Executive Summary

Cities already record thousands of traffic incidents: vehicle breakdowns, potholes, water logging, rallies, processions, VIP movements, construction closures, and accidents. But most of that history is used only for reporting after the fact.

ATLAS changes that.

ATLAS reads Bengaluru's historical Astram incident data and turns it into a live operational command system. It learns when the city gets busy, which areas trigger follow-on congestion, where drainage-sensitive zones exist, which police stations are overloaded, and how officers should be deployed when a new event is planned or detected.

The result is not just a dashboard. It is a decision-support prototype that answers:

- If an event happens here, how severe will it be?
- Where will congestion spread next?
- Which police stations should respond?
- How many officers are needed?
- Where should barricades and diversions be placed?
- How much evidence supports this prediction?
- Did the system actually work when replayed against historical data?

ATLAS is designed around one principle: **trustworthy prediction, not confident guessing**. If the data is strong, it makes a prediction. If the data is thin, it says so.

2. The Problem

Traffic operations teams face three recurring problems:

1. Event impact is not quantified early enough.

A rally, sports event, VIP movement, or road closure may be known in advance, but its likely traffic impact is often judged from experience rather than data.

2. Congestion spreads, but response is local.

One breakdown or water-logging point can trigger downstream congestion elsewhere. Existing systems often treat incidents as isolated dots.

3. Historical incident data is underused.

Cities collect incident logs, but these records rarely become predictive intelligence. They become reports, not operational foresight.

ATLAS addresses these gaps by transforming past incident history into a system that can forecast, explain, and recommend action.

3. What ATLAS Does

ATLAS has six main capabilities.

3.1 Behavioral Fingerprint

ATLAS learns the city's traffic rhythm from timestamps alone.

It identifies:

- Which hours are normally busy.
- Which weekday patterns repeat.
- Which days are abnormal surge days.
- Whether the city has a learnable rhythm.

Example: ATLAS flagged **8 anomaly days** without being given any festival calendar.

That means the system discovered unusual city-wide traffic days purely from the incident data.

3.2 Cascade Graph

ATLAS learns how incidents spread.

It examines every historical incident and asks:

If something happened at station A, did another incident appear at station B within the next hour and within nearby geography?

When this happens repeatedly and significantly more than random chance, ATLAS records a cascade link.

Example learned links:

From	To	Meaning
Pulikeshinagar	High Ground	Trouble at Pulikeshinagar often precedes trouble near High Ground
Adugodu	Madiwala	A likely spillover route in southern traffic movement
High Ground	Cubbon Park	Central-area issues frequently propagate nearby

This lets ATLAS predict secondary impact zones after a new event is dropped on the map.

3.3 Event Severity Index (ESI)

ESI is a 0-10 score that summarizes operational severity.

It combines:

- Event kind.
- Expected size.
- Road closure requirement.
- Predicted duration.
- Historical cascade risk.
- Corridor load.
- Local police station context.
- Evidence confidence.

Most importantly, ATLAS shows the breakdown behind the score. It is not a black box.

3.4 Optimal Deployment

ATLAS recommends which police stations should respond to each predicted hotspot.

It considers:

- Distance from station to event.
- Current station load by hour.
- Primary event location.
- Predicted secondary cascade locations.
- Officer requirements.

It then uses the **Hungarian assignment algorithm** to find the mathematically best station-to-location assignment for minimum expected response time.

3.5 Time-Machine Replay

ATLAS can replay any historical day as a map time-lapse.

A day is played minute-by-minute using the original timestamps from the CSV. Each incident appears when it actually happened.

This allows reviewers to see the real city history unfold, not a fabricated simulation.

3.6 Evidence-Gated Forecasting

ATLAS does not pretend every prediction is equally reliable.

For each event type, it checks how many similar historical incidents exist.

- Many examples -> data-driven prediction.
- Some examples -> limited evidence warning.
- Very few examples -> insufficient evidence warning.

This is critical for trust. A system that admits uncertainty is safer than one that overclaims.

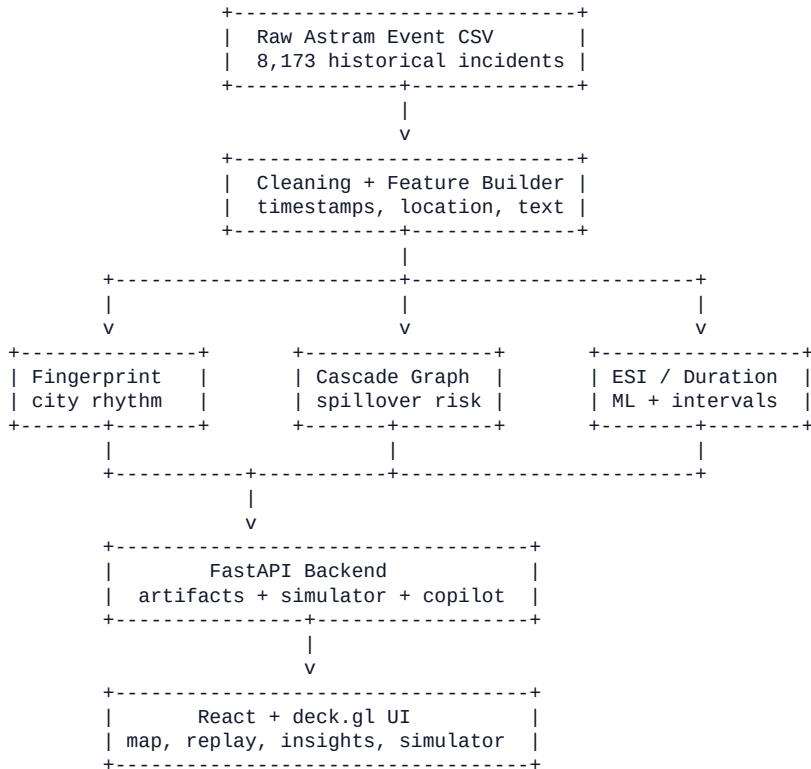
4. What Makes ATLAS Different

Most hackathon traffic projects become dashboards. ATLAS goes further.

Typical dashboard	ATLAS
Shows past incidents	Replays past and predicts future hotspots
Shows heatmaps	Learns city rhythm and anomaly days
Counts incidents	Finds statistically significant cascade links
Shows a map	Lets users drop hypothetical events and get a response plan
Gives charts	Recommends officer deployment, barricades, and diversions
Hides uncertainty	Shows confidence and evidence base

ATLAS is not just visualization. It is a prototype for operational decision-making.

5. System Overview



6. Dataset Understanding

The raw dataset contains **8,173 rows** and **46 columns**.

After cleaning and filtering Bengaluru coordinates, ATLAS uses **8,041 valid incidents**.

6.1 Important fields

Field	Meaning	Use in ATLAS
start_datetime	When the incident started	Replay, fingerprint, cascade, model features
closed_datetime	When the incident was closed	Duration target for ML
latitude, longitude	Incident location	Map, distance, nearest station, corridor logic
event_cause	Incident category	Severity, evidence, semantic archetypes
police_station	Station associated with incident	Cascade graph, deployment
corridor	Named traffic corridor	Corridor risk and map lines
zone	Police/geographic zone	Drainage Vulnerability Index and audit
description	Incident text, including Kannada rows	Text flags and multilingual archetypes
requires_road_closure	Whether closure was required	ESI and barricading
priority	Operational priority	ML feature

Field	Meaning	Use in ATLAS
veh_type, veh_no, cargo_material	Vehicle-related context	Breakdown and severity features

6.2 Important data reality

The data is messy, which is normal for real civic systems.

Key observations:

- 60% of incidents are vehicle_breakdown.
- 94% of rows are unplanned events.
- Only around one-third have reliable close-out times.
- Some tickets were closed days later, causing misleading average durations.
- Around 13% of descriptions contain Kannada text.

ATLAS handles these realities by:

- Using medians instead of means for skewed duration summaries.
- Training only on rows with trustworthy duration labels.
- Showing evidence warnings when data is thin.
- Using language-agnostic character n-gram analysis for Kannada text.

7. Algorithms and Computations

This section explains the system without assuming machine-learning background.

7.1 Cleaning and Feature Engineering

ATLAS first converts the raw CSV into a reliable table.

It computes:

- duration_min = closed time minus start time.
- response_min = created time minus start time.
- Hour of day, weekday, month, weekend flag.
- Weather/water-related text flags.
- Event-related text flags.
- Road-closure flag.
- Vehicle and cargo indicators.
- Route-path length indicators.
- Officer assignment indicators.

This turns a messy operational log into structured signals that models can use.

7.2 Behavioral Fingerprint

ATLAS counts incidents by hour and weekday to learn the city's rhythm.

It creates a 24-hour breathing pattern:

```
00h 01h 02h ... 08h 09h 10h ...
low low low    high high high
```

It also detects unusual days using a robust anomaly method:

normal day: 40-60 incidents
anomaly day: 100+ incidents
major surge: 200+ incidents

Example anomaly days detected:

Date	Incidents	Interpretation
2024-03-07	214	Major city-wide surge day
2023-12-16	200	Major city-wide surge day
2024-04-06	141	Strong anomaly

The important point: ATLAS discovers these without a festival calendar.

7.3 Cascade Graph

ATLAS learns incident spread patterns.

For every incident, it checks the next 60 minutes and nearby geography.

If incidents repeatedly move from station A to station B, ATLAS learns a link:

Incident at A ---> incident at B shortly after

But ATLAS does not trust every link. It checks whether the pattern is stronger than random chance.

It uses:

- Co-occurrence counting.
- Distance filtering.
- Time-decay weighting.
- Lift over random baseline.
- Permutation testing.

Example:

Pulikeshinagar -> High Ground
probability: 0.69
lift: 30.7x
expected delay: 19 minutes

Plain meaning:

When Pulikeshinagar has trouble, High Ground is much more likely than normal to see trouble shortly after.

7.4 Event Severity Index

ESI is a 0-10 operational score.

It uses two kinds of information:

1. **Event inputs**

- Event kind.
- Size.
- Duration.
- Road closure.

2. **Historical signals**

- Predicted duration from ML.
- Cascade risk near the station.
- Corridor load.
- Evidence count.

Simplified formula:

ESI = event risk + closure risk + size risk
+ duration risk + cascade risk + corridor risk

The UI shows the breakdown, so reviewers can see exactly why a score was assigned.

Important honesty feature:

If ATLAS has only a small number of examples for an event type, it labels the result as limited evidence.

7.5 Duration Prediction Model

The ML model predicts event duration.

ATLAS uses:

- Gradient Boosting Regression.
- Quantile regression heads for lower and upper uncertainty bands.
- Out-of-time validation.

Because many tickets were never closed, only **2,380 rows** have reliable duration labels and are used for training.

This is intentional. The system does not fabricate labels for unclosed tickets.

Validation snapshot:

Metric	Value
Holdout rows	476
MAE	35.39 min
90% interval coverage	88.4%
Avg interval width	151 min

Plain meaning:

The point estimate is noisy, but the uncertainty range is honest and calibrated.

7.6 Optimal Deployment

When an event is dropped, ATLAS predicts:

- the main event location,
- likely cascade locations,
- stations that can respond,
- expected travel time,
- station load.

It builds a matrix:

	Event	Hotspot 1	Hotspot 2	Hotspot 3
Station A	6 min	12 min	15 min	10 min
Station B	8 min	5 min	7 min	11 min
Station C	14 min	6 min	4 min	5 min

Then it solves the assignment problem:
Which station should cover which location to minimize total response time?
This is solved using the Hungarian algorithm.

7.7 Time-Machine Backtest

This is the proof system.
ATLAS replays real historical days.
For each day:
1. It looks only at the morning.
2. It predicts which police stations will become hotspots in the afternoon.
3. It compares the prediction to what actually happened.

Result:

Method	New afternoon hotspot recall
Persistence baseline	31.2%
ATLAS	53.2%

Plain meaning:
ATLAS catches surprise afternoon hotspots better than simply assuming the afternoon will look like the morning.

7.8 Multilingual Archetypes

ATLAS found that around 13% of descriptions are in Kannada.
Instead of ignoring those rows, it uses character n-gram TF-IDF.
This means it looks at small pieces of words or characters, not full English words. That allows it to cluster Kannada and English complaint text without translation.
Example capability:
Kannada drainage complaints + English drainage complaints
-> grouped into similar incident archetypes
This makes the system more inclusive and locally relevant.

8. Product Screens

8.1 Ops Tab

Used for live simulation.
User can:
• Search or click a location.
• Pick event kind.
• Pick size.
• Set duration.
• Mark road closure.

- Run simulation.

ATLAS returns:

- ESI score.
- Evidence tier.
- Predicted cascade locations.
- Officer deployment.
- Barricades.
- Diversions.

8.2 Replay Tab

Used to replay a real historical day.

User can:

- Pick anomaly date.
- Press play.
- Watch incidents appear by timestamp.
- See cascade edges among active stations.

This proves that the prototype uses real data, not fabricated animation.

8.3 Insights Tab

Used for proof and explanation.

Shows:

- Time-machine backtest.
- New hotspot discovery performance.
- Multilingual archetypes.
- Evidence counts by event cause.
- Out-of-time validation.

8.4 Fingerprint Tab

Used to show city rhythm.

Shows:

- 24-hour breathing pattern.
- 8 anomaly days found without a calendar.
- Drainage Vulnerability Index by zone.

8.5 Audit Tab

Used to estimate historical operational impact.

Shows:

- Officer-hours saved.
- Average response improvement.

- P90 response improvement.
- Zone-level breakdown.
- Fairness/Gini note.

8.6 Co-pilot Tab

Used for grounded question answering.

It answers questions like:

- What happens in Yelahanka at 9 AM?
- Which corridor has high event load?
- What is the top cause in a station?

It answers only from the dataset and shows row counts. If there is no data, it refuses.

9. Impact

ATLAS has three categories of impact.

9.1 Operational impact

- Predicts surprise hotspots before they emerge.
- Recommends station deployment instead of relying only on experience.
- Identifies cascade spillover zones.
- Supports barricade and diversion planning.

9.2 Strategic impact

- Converts raw incident logs into reusable intelligence.
- Provides a learning loop after every event.
- Helps cities compare actual response vs data-driven response.
- Builds evidence for resource allocation.

9.3 Inclusion impact

- Kannada complaints are included in text clustering.
- The system does not ignore local-language incident descriptions.
- This makes the intelligence layer more representative of real citizen and field-reporter inputs.

10. Current Verified Metrics

Metric	Value
Raw incidents	8,173
Cleaned Bengaluru incidents	8,041
Police station nodes	54
Cascade links	466
Corridors profiled	21

Metric	Value
Anomaly days detected	8
Kannada descriptions clustered	856
Time-machine days evaluated	151
Surprise hotspot recall: baseline	31.2%
Surprise hotspot recall: ATLAS	53.2%
Officer-hours saved in audit	78.4
Duration model MAE	35.39 min
90% interval coverage	88.4%

11. How This Can Scale

ATLAS is designed to be city-portable.

11.1 Scaling to more data

If given more months or years of incident history, ATLAS improves in three ways:

- Better duration prediction.
- Stronger cascade links.
- More reliable evidence for rare event types.

11.2 Scaling to live operations

The current prototype is batch-built from a CSV. In production, the same architecture can ingest live events.

```
Live incident stream
  -> rolling feature store
  -> updated cascade risk
  -> live deployment recommendations
  -> post-event learning
```

11.3 Scaling to multiple cities

Because ATLAS only needs a standard incident log, it can be reused in other cities.

Required fields:

- timestamp,
- latitude/longitude,
- incident cause,
- station/zone,
- optional closure and resolution times.

No sensors or expensive new hardware are required.

11.4 Scaling to monsoon operations

With monsoon-season data, the Drainage Vulnerability Index becomes a strong flood-response tool.

It can help decide:

- where to pre-position pumps,

- where to keep traffic officers during storms,
- which corridors need diversion plans,
- which zones repeatedly suffer water-logging.

11.5 Scaling to command-center deployment

Future production modules:

- live feed ingestion,
- user authentication,
- officer roster integration,
- map layers for road closures,
- SMS/WhatsApp alerts,
- station workload dashboard,
- post-event learning loop.

12. Limitations and Honesty

ATLAS is intentionally transparent about its limits.

12.1 Limited planning-event data

The dataset has many vehicle breakdowns but fewer rallies, sports matches, VIP movements, and protests.

So ATLAS labels such predictions as limited evidence.

12.2 Missing close times

Many incidents were never formally closed in the ticketing system.

Therefore, ATLAS trains duration models only on rows with reliable closure times.

12.3 No monsoon season in the dataset window

The dataset window is mostly dry-season. DVI is therefore a method demonstration, not a full monsoon model yet.

12.4 Approximate response modelling

The audit estimates response time from distance and station load. It does not yet use live GPS travel times.

12.5 UI is a prototype

The frontend is designed for demo and evaluation, not production security or mobile deployment.

13. Why This Matters

ATLAS demonstrates a practical idea:

Cities do not always need new sensors to become smarter. Sometimes they already have the data - they just need the intelligence layer.

By turning historical incident records into prediction, replay, deployment, and audit, ATLAS shows how a city can move from reactive traffic response to evidence-driven traffic command.

14. One-Minute Pitch

Bengaluru already records thousands of traffic incidents, but that data is mostly used after the fact. ATLAS turns it into a live prediction system.

It learns the city's traffic rhythm, finds abnormal surge days without a calendar, detects how congestion cascades from one police station area to another, and recommends the fastest officer deployment using mathematical optimization.

Most importantly, we proved it on history. Replaying 151 real days, ATLAS caught 53% of surprise afternoon hotspots, compared with 31% for a persistence baseline. It also reads Kannada and English incident descriptions, includes confidence intervals, and warns when evidence is thin.

ATLAS is not just a map. It is a prototype for a city-scale traffic command system that learns from every incident and helps police act before congestion spreads.

15. Conclusion

ATLAS is a working prototype of a predictive traffic command system. It is built from one anonymized CSV, runs locally, produces explainable recommendations, and is honest about uncertainty.

It combines:

- historical replay,
- cascade prediction,
- multilingual text analysis,
- evidence-gated severity scoring,
- optimal officer deployment,
- and counterfactual audit.

The result is a system that helps city traffic teams move from reactive response to proactive, evidence-backed action.

Project: ATLAS - Atram-Trained Live-Adaptive System **Theme:** Event-Driven Congestion, Planned & Unplanned

Prototype status: Working full-stack local demo **Last updated:** 2026-06-21